CCCA Cell-State Trajectory — Zhang et al., 2019

Follicular Lymphoma · 10x · 5,645 malignant B cells (trajectory scoped to 5,449 non-cycling)

GO / NO-GO GATES

Gate 1 Integration & continuity — **GO**

Patients/samples intermix on one continuous manifold (iLISI 1.00→2.26 of 4 at Harmony θ=10; patient 1.67/2); single connected component, 0 detached islands. Strongly clonal FL batch integrates.

Gate 2 Module geography — **GO**

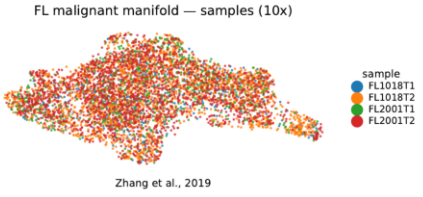
Non-branching: on the cycling-scoped Body, Gast/pSom co-vary (r=+0.33) and both oppose Crest (Crest–Gast −0.83, Crest–pSom −0.41); Crest is the terminal readout (high-decile pt=0.58). Single collapsed developmental axis.

Gate 3 Cross-repression (branching only) — **NA**

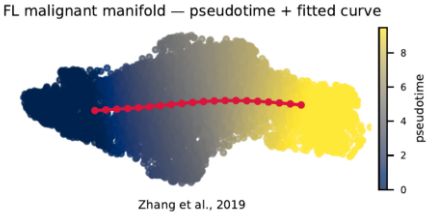
No Gast/pSom two-arm bifurcation (curve: 2 tips, 0 forks) — branching-only correlation/modules steps not produced.

STEP FIGURES

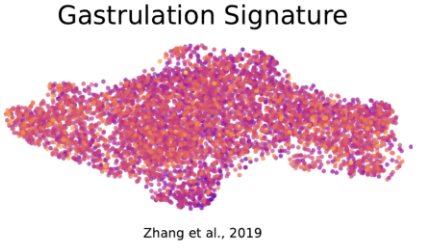
Gate 1 — sample integration



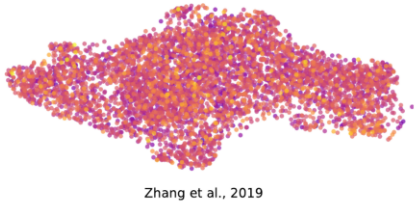
Trajectory — pseudotime + curve



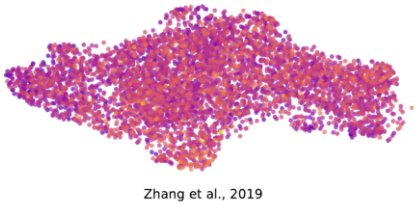
Gastrulation



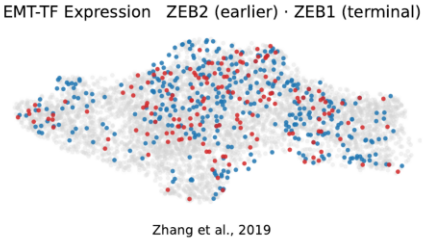
Somitogenesis



Neural Crest



EMT-TFs (ZEB2/ZEB1)



PARAMETER CHOICES & RATIONALE

Sample threshold	≥100 floor — kept all 4 samples (largest gap ratio 2.74 not a clean cliff; min 680 is a genuine primary FL tumour)
Integration	Harmony θ=10 on sample (chosen on iLISI sweep + embedding; FL clones are patient-private, need strong θ)
Embedding	n_pcs=20, n_neighbors=30, UMAP spread=1.3 (10x precedents); 636 overdispersed genes
Cycling scoping	Excluded a 196-cell, 95%-cycling lobe where pSom was proliferation-confounded (pSom–cycling r +0.61→+0.18)
Geometry	Linear / non-branching — scFates curve (ElPiGraph) Nodes=20, λ=0.3, μ=100; 2 tips 0 forks; fold-free (self-approach 2.96, DC1 +0.94)
Root	Curve tip at the Gast/pSom-high, Crest-low developmental origin (Gast↓ rho −0.41, Crest↑ rho +0.43 along pt)
EMT-TFs	ZEB2 (28% expr, earlier) · ZEB1 (12%, most terminal); SNAI/TWIST/FOXC/PRRX below 3% detection; VIM excluded (marker not TF)
Module means (raw)	Gast +0.012 · pSom +0.005 · Crest +0.010

VERDICT

✓ **Pattern reproduces (non-branching). A single Gast→Crest developmental axis: gastrulation/somitogenesis** programs at the root, neural-crest as the terminal readout. No competing arms — Gate 3 not applicable.

Hub: n/a (non-branching, no convergence zone / cross-repression arm). Hub r: n/a.